

Solution Requirements (Functional & Non-functional)

Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Dashboard Visualization	Display an interactive Tableau dashboard with electricity consumption KPIs such as Total Usage, Average Usage, Highest Usage, and Lowest Usage.
FR-2	State-wise Electricity Analysis	Analyze electricity consumption across different states and compare state-wise usage patterns.
FR-3	Region-wise Electricity Analysis	Compare electricity usage among Northern, Southern, Eastern, Western, and Central regions.
FR-4	Monthly Trend Analysis	Display monthly electricity consumption trends and identify seasonal demand variations.
FR-5	Geographic Map Analysis	Display electricity consumption on an India map using state locations for better geographic understanding.
FR-6	Top 10 Electricity Consumers	Identify and display the top ten highest electricity-consuming states using interactive charts.
FR-7	Interactive Filters	Allow users to filter dashboard data by State, Region, and Date for customized analysis.
FR-8	Tableau Story	Present a guided Tableau Story explaining electricity consumption trends, regional comparisons, and business insights.
FR-9	Flask Web Application	Embed the Tableau Dashboard and Tableau Story into a Flask web application for easy browser access.
FR-10	Business Insights	Generate summarized insights and conclusions to support electricity boards and government decision-makers in energy planning.

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be simple to use with clear navigation, readable charts, and interactive filters for all users.
NFR-2	Security	The Flask application should securely present electricity analysis without exposing or modifying the original dataset.
NFR-3	Reliability	The dashboard should consistently display accurate calculations and visualizations based on the electricity consumption dataset.
NFR-4	Performance	Dashboard pages, filters, and visualizations should load quickly and respond smoothly to user interactions.
NFR-5	Availability	The Tableau Public dashboard and Flask web application should be available whenever users need to access electricity analysis.
NFR-6	Scalability	The system should support larger electricity datasets, additional dashboards, and future analytical features without major redesign.
